

Cleft lip and palate (CLP) is a split in the upper lip and/or the roof of the mouth. CLP is present from birth.

CLP can be caused by various genetic and environmental factors.

CDH1 is a gene involved in lip and mouth development in a fetus before a baby is born.

Scientists hypothesised that CLP is caused by a 'two-hit' model:

- a 'genetic hit' through loss of *CDH1* function
- an 'environmental hit' leading to *CDH1* methylation caused when inflammation affects a fetus.

Inflammation can result from the mother having an infection or diabetes.

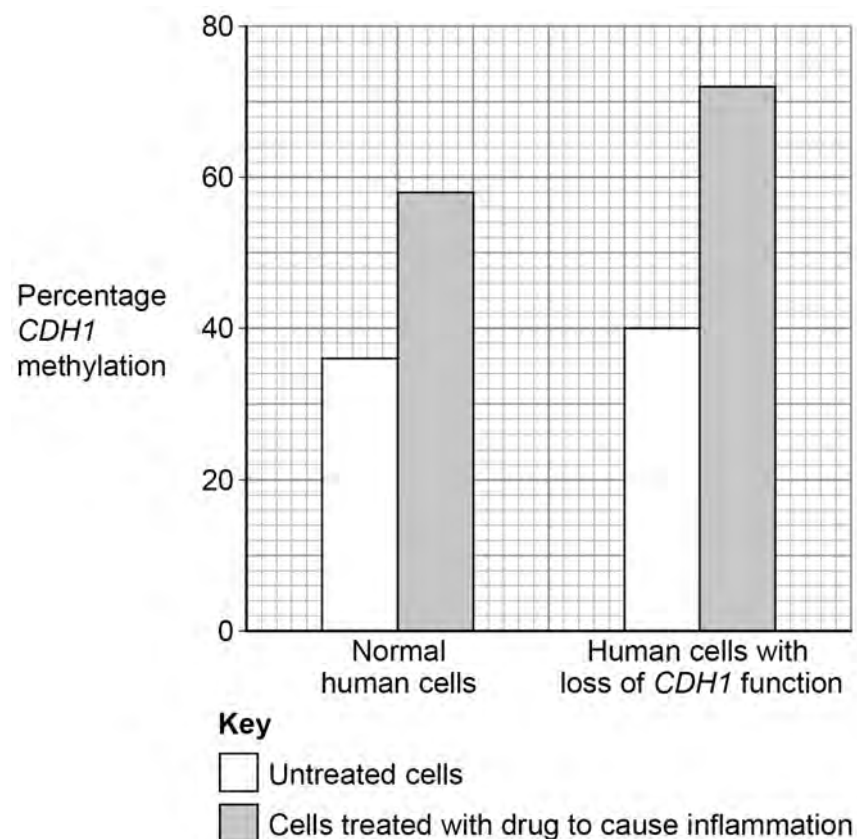
Scientists investigated the 'two-hit' model using human cells grown *in vitro*.

They:

- grew normal cells and cells with a loss of *CDH1* function
- treated some of each type of cell with a drug to cause inflammation and left some of each type of cell untreated
- measured the percentage *CDH1* methylation of the cells.

Figure 4 shows the scientists' results.

Figure 4



05.1

The frequency of live births with CLP is 1 in 700

Despite knowing this frequency, the Hardy–Weinberg equation **cannot** be used to estimate **any** allele frequencies for CLP.

Suggest **two** reasons why.

[2 marks]

1 _____

2 _____

05.2

Describe **one** effect methylation could have on *CDH1*

[1 mark]

05.3

Suggest **and** explain **three** reasons why the data in **Figure 4** might **not** provide evidence for the ‘two-hit’ model.

[3 marks]

1 _____

2 _____

3 _____
